VEX IQ / ROBOTICS COMPONENTS

The modular VEX IQ platform & the 2nd-generation (2nd gen) system — structure, motion, gears, motors, sensors, electronics, programming and drivetrains on one sheet. VEX IQ 2nd gen, Education & Competition Kits, VIQRC.

1 : 2	speed : torque — small gear driving a 2× gear doubles torque, halves speed
2nd Gen	colour-screen IQ Brain + Smart Motors with built-in encoders at 40 Hz control
12 & 60 T	gears & sprockets — a 5 : 1 range to teach tooth-count ratio maths
1× & 2×	square & cross shafts, 0.25×·1×·4×·6× pitch sizes, rubber collars + hubs
Plastic + Al	tool-free modular snap structure (IQ) · same grid geometry in aluminium (V5)

THE SYSTEM

THE VEX IQ SYSTEM — WHAT IS IN THE KIT

2nd-generation kits: IQ Education Kit 228-8899 (~1,052 parts) and IQ Competition Kit 228-7980 (~1,773 parts). Tool-free, Grades 5–8.

VEX IQ is a **modular, tool-free plastic robotics platform**: strong snap-together structural pieces hold **smart electronics** that you program in VEXcode (Blocks, Python or C++). Every subsystem is a category of component you mix, match and reuse build after build:

Subsystem	2nd-gen kit parts (examples)	What it does for your robot
Structural	Beams 1× & 2×, C-Channels, U-Channels, Plates, angled beams (30°/45°/60°/90°, Tee), corner connectors	The frame, arms and chassis skeleton — snap together in an exact grid
Fasteners	Connector Pins (1×, 2×, 3×), standoffs (0.25×–6× pitch), standoff connectors, pitch spacers, Pin Tool	Join beams at right angles, space plates apart, hold sub-assemblies firmly
Gears / sprockets / chain	Spur gears 12T, 24T, 36T, 48T + 23T crank arm; sprockets 8T, 16T, 24T; chain links; rack	Change speed vs torque, transmit rotation, drive linear racks
Wheels & treads	200 mm travel wheel & tire, 200 mm omni wheel, balloon tire, wheel hubs, traction links, tank tread, wide spool	Move, steer, climb, grip and drive the robot across the field
Shafts & collars	Cross/square shafts (0.32×, 1×, 2×, 4×, 5×, 6× pitch), motor shafts, rubber shaft collars, washers, spacers	Transmit rotation from motor to gears to wheels
Motors	Smart Motor (2nd gen) — 4 in Education Kit, 6 in Competition Kit	Drive wheels, lifts and mechanisms with built-in encoder feedback
Sensors	Distance, Optical, Bumper, Touch LED (2nd gen); Inertial/gyro built into the Brain; Vision Sensor add-on	Let the robot see, feel, measure and react to its world
Electronics	IQ Robot Brain (2nd gen), IQ Controller (2nd gen), 7.2 V Robot Battery, Smart Cables, USB-C charger	The brain, hands, eyes, power and remote that tie it all together
Field / competition	VIQRC field, foam field tiles, competition elements (game objects), Robot Skills & Teamwork matches	A standard arena every team builds for — the competition ecosystem

STRUCTURE

STRUCTURE — THE MODULAR TOOLKIT

Structure is every non-moving part that shapes the robot: the frame, arms, mounts and braced skeleton. VEX IQ uses **snap-in connector pins and standoffs**, so no screws or tools are needed.

The beams — modular toolkit

- **Beams 1×1 / 1×N** — the basic square cross-section bar, in lengths 1×, 2×, 4×…20×; the 1× series is the classic beam
- **C-Channels** — a beam with one open side (a 'C' in section), giving two parallel walls to nest gears and shafts inside
- **U-Channels** — an open 'U' section; rooms for flanges, stiffening and holding wheels on each face
- **Plates** — flat sheets (2×8, 4×8, 4×4, 3×3) to cover and connect areas into a rigid deck
- **Angled & Tee beams** — 30°, 45°, 60°, 90° (right-angle) and Tee beams to change direction of the frame
- **Corner connectors & gussets** — lock two or three beams together at right angles

Materials & hole pattern

IQ structure is tough, light **ABS-style plastic** (safe for classrooms, no sharp metal edges, floats of colour per length). The big sibling **V5 uses aluminium** with the same idea. Holes follow the **standard VEX grid pattern**: they line up on a regular square lattice every 1× pitch along each axis, with the odd hole offset so you can also join on the half-grid. Because the hole spacing is identical on every beam, any two beams can bolt/pin together at any overlapping hole — that is what makes the build truly modular.

Rule of thumb: design the chassis first (strength + weight), then mount motors and electronics, then brace anything that flexes. Plastic bends a little under load — triangles and cross-bracing stop twist.

MOTION

MOTION COMPONENTS

Motion components take the motor's spin and turn it into movement, at the speed and torque your robot needs.

Gears & sprockets

- **Spur gears** — flat, toothed wheels on parallel shafts; the workhorse for changing speed/torque
- **Bevel & crown gears** — 45° teeth that change the axis of rotation by 90°
- **Sprockets + chain** — toothed wheels joined by chain links; drive shafts that are too far apart for gears to touch
- **Rack & pinion** — a straight toothed 'rack' pushed back & forth by a gear 'pinion' — for linear motion (slides, lifts)

Shafts, hubs & collars

- **Shafts** — cross/square cross-section rods (0.25×, 1×, 2×, 4×, 5×, 6× pitch) that carry rotation; IQ uses square/cross so gears can't slip around them
- **Wheel hubs** — lock a wheel to a shaft (e.g. Wheel Hub 44 mm / 48.5 mm wide hub)
- **Rubber shaft collars & spacers** — stop shafts sliding side-to-side and set lateral position

Wheels & treads

- **Omni wheel** — rollers around the rim let it roll sideways; gives strafing and smooth turns
- **Traction / travel wheel** — high-grip rubber tire (e.g. 200 mm travel) for climbing and pushing
- **Gear wheel / balloon tire** — big-diameter wheels that climb over field objects
- **Tank tread** — continuous rubber/pivot tread on sprockets for steep inclines and rough ground

Speed vs torque relationship — a **1:1** pair does nothing but turn the direction/axis; any other ratio trades **speed for torque** (see Gearing). Small gear driving a big gear = more torque, less speed.

GEARING

GEARS & GEAR RATIOS

A gear's **tooth count** sets how fast and how hard it turns. Teeth must mesh 1-for-1, so the **ratio** of two meshed gears equals their tooth counts.

The gear-ratio rule

- **Gear ratio = driven ÷ driver** (output gear teeth ÷ input/driver gear teeth)
- **Ratio > 1** → speed reduction, **torque multiplication** (force up, rpm down)
- **Ratio < 1** → speed increase, torque reduction (rpm up, force down)
- **1 : 1** → same speed/torque, just transmits (or reverses) the motion

Worked examples

- **12T driving 36T** — 36 ÷ 12 = **3 : 1** → output spins 3× slower but delivers **3× the torque** (perfect for a claw or lift arm)
- **36T driving 12T** — 12 ÷ 36 = **1 : 3** → output spins 3× faster with 1/3 the torque (drivetrain top speed)
- **12T : 60T** — 60 ÷ 12 = **5 : 1**, the full poster range: big torque for heavy mechanisms

Idler gears

An **idler gear** sits between driver and driven but is not connected to a load or motor. It only **reverses direction** and keeps teeth meshed — it does **NOT** change the overall ratio. Two gears spin in opposite directions; three gears spin the first and last in the same direction.

Chain & sprocket maths is identical: sprocket tooth counts follow the same driven ÷ driver rule, so long runs of chain can step speed and torque just like direct gears. Gear trains in series multiply — two 3:1 stages give 9:1 torque.

MOTORS

MOTORS & POWER

The **Smart Motor (2nd gen)** is more than a motor — it is a self-contained smart device. A built-in **Texas Instruments MSP430 processor (16 MHz)** controls it via an H-Bridge and runs **PID loops internally** to hold your commanded speed and position.

Spec & feedback

- **Free speed** ≈ **120 rpm** (output shaft), stall torque ≈ **0.414 N·m**, peak power ≈ 1.4 W continuous
- **Built-in quadrature encoder** — 960 ticks/rev, position reported to **0.375°**
- Reports **velocity, position, current and torque** live to the Brain dashboard
- Software **torque limit** and consistent ±1% performance regardless of battery charge

Motor ports & setup

- Motors plug into the Brain's **smart ports**; the port auto-detects the device (motor or sensor) — no configuration needed
- You set direction, speed, acceleration, position and a **max torque** in code: *motor.setVelocity(50, PERCENT)*

Torque vs speed & gearing

A bare motor is fast but weak. **Gear it down** (small pinion → big gear) to multiply torque for lifting, or gear up for a fast drivetrain. The Smart Motor draws more current the harder it works.

Bumper / limit caution: a **stalled** motor (blocked from turning while powered) draws high current and heats up. Use **Bumper switches or limit switches** to stop a mechanism at its travel ends, keep mechanisms mechanically free, and never leave a robot stalled under power match after match.

SENSORS

SENSORS

Sensors feed the Brain data about the robot and its environment, so programs can make decisions. All plug into the same smart ports as motors.

2nd-gen sensor set

- **Distance Sensor** — measures range to objects (ultrasonic, in cm); use to approach, stop before walls, or detect objects to pick up
- **Optical / Colour Sensor** — detects colour, brightness and gesture; use to line-follow (colour signature) or signal-match game objects
- **Bumper Switch** — a physical push button (digital on/off); use as a limit/contact sensor to stop a drive or a mechanism
- **Touch LED** — an RGB light that is also a push button; use as a button, and light it green/red to signal state
- **Inertial Sensor** — 6-axis gyro + accelerometer; the 2nd-gen Brain has one **built-in**; use the gyro to turn exact angles and drive straight
- **Vision Sensor** — a camera (add-on) that detects objects/colour signatures in its field of view; use for object tracking and targeting

What each measures → how to use it

Sensor	Measures	Typical use
Distance	range (cm) to an object	stop before a wall; sense a game piece ahea
Optical/colour	colour, brightness	line following, colour sorting
Bumper	pressed / released	limit switch, contact detection
Touch LED	tap, plus RGB output	start button + status light
Inertial/gyro	yaw, pitch, roll, acceleration	turn exactly 90°, drive straight, balance
Vision	objects & colours in frame	track and aim at game objects

Combine sensors with **if/else logic** and **PID** (see Programming) to make the robot react, not just repeat.

ELECTRONICS

ELECTRONICS — BRAIN, POWER & REMOTE

The **IQ Robot Brain (2nd gen)** is the heart of the robot: it runs programs, reads every sensor and motor, and talks to the remote.

IQ Brain (2nd gen)

- **Colour touchscreen** interface with multilingual menu + dashboards (live diagnostics on every port)
- **12 smart ports** — motors and sensors plug in interchangeably and are **auto-detected**
- **Built-in 6-axis Inertial sensor** (gyro + accelerometer) and **Bluetooth 5.0** radio (built-in → no separate radio card needed)
- Roughly **4× CPU, 12× RAM, 32× flash** vs the 1st-gen Brain → it runs **Python and C++** in VEXcode, not just Blocks
- MicroSD slot for storage & datalogging; 8 user program slots; automatic device/controller updates

Power & energy

- **Robot Battery (2nd gen): 7.2 V, 2,000 mAh** lithium-ion (recent) / NiMH (earlier) — rechargeable via USB-C
- On-board **LED battery-level indicator**; lithium cells hold voltage longer so motors don't weaken at end of charge

Remote & field

- **IQ Controller (2nd gen)** — wireless with two joysticks + buttons; Bluetooth 5.0 pairing is menu-driven (no cable); it can **wirelessly download programs** to the Brain
- Competition control: programs run in **Autonomous** or **Driver Control** mode; the Brain auto-tracks match timers; an **E-stop / field-control** stops the robot safely at the end

DESIGN & BUILD

THE DESIGN & BUILD PROCESS

A strong VEX IQ robot is designed before it is built. Follow this loop to turn an idea into a working, competition-ready machine.

1. Brainstorm
Understand the game/field; list the tasks, constraints (size limit, part budget, time) and ideas. Sketch at least 2–3 concept options.

2. Draw / diagram
Put your best idea on paper or in VEXcode VR / CAD: top and side views, where the motors, wheels, lift and Brain go, how cables route.

3. Build the chassis
The rigid base frame first — beams and corner connectors, sized to house the Brain, battery and motors. Keep it light and square.

4. Build the drivetrain
Motors, gears/sprockets and wheels from Motion & Gearing. Choose the gear ratio for the speed vs torque your game needs.

5. Build the mechanism
Arm, lift or claw — gear it for torque; brace it; add limit/bumper switches at travel ends so it can't jam.

6. Program
Drive motors, read sensors, test each action in small steps (see Programming).

7. Test & iterate
Run on the field, measure, break things on purpose, then improve. **Iteration is the whole point** — every redesign makes it better.

Golden rule: prototype ruthlessly and let the robot tell you what to fix — the design loop is never finished.

PROGRAMMING

PROGRAMMING — VEXCODE

The 2nd-gen Brain is programmed in **VEXcode IQ**, which offers three ways to write the same logic.

Blocks vs Python

- **Blocks** — drag-and-drop visual coding; perfect for beginners; compiles to the same behaviour
- **Python** — text code on the 2nd-gen Brain; the step up in rigor (also **C++** for advanced teams)
- Same devices, sensors and logic in every language — start in Blocks, graduate to Python

Key drivers & sensor readings

- Motor: *motor.setVelocity(60, PERCENT)*, *motor.spin(FORWARD)*, *motor.stop()*, *motor.setPosition(0, DEGREES)*, *motor.velocity()*
- Sensors: *distance.getDistance(MM)*, *optical.color()*, *bumper.pressing()*, *inertial.rotation()*
- Control flow: *if/else*, *while True* loops, *forever*, event-driven 'when pressed' handlers

PID — make it exact

PID (Proportional–Integral–Derivative) is a control loop: it compares a *goal* (target angle or line) to the *reading* and adjusts motor power by the error. A simple **P** term gets you most of the way: the bigger the error, the harder it corrects. Use PID to **drive straight** (gyro) and to **line-follow** (optical colour) smoothly instead of with jerky on/off.

Auton vs Driver Control

- **Autonomous (Auton)** — the robot drives itself from a program: start button → run exact moves → stop. Scores during the autonomous period.
- **Driver Control** — the driver steers live with the Controller joysticks (tank or arcade controls).
- Strong teams nail a reliable auton **and** a smooth driver-control loop for the full match.

DRIVETRAINS

DRIVETRAINS

The drivetrain is the wheel-and-motor system that moves the whole robot. Layout choice decides speed, turning and traction.

Common drivetrains

- **4-wheel drive** — four driven wheels, simple and reliable; good all-round balance
- **6-wheel drop ('drop centre')** — two extra centre wheels set lower; excellent over bumps and field objects, common in VIQRC
- **Omni drive** — omni wheels allow **strafing** (slide sideways) and instant direction changes
- **X-drive** — four omni wheels angled 45°; the whole robot is an omni-drivetrain for maximum mobility in tight spaces
- **Tank (skid-steer) drive** — left and right sides driven independently; turns by driving one side faster than the other

Turning: skid vs point-turn

A **skid turn** is a wide arc — one side drives while the other slows or idles; gentle on the robot. A **point turn (spin)** drives the two sides in **opposite directions** so the robot rotates about its centre, footprint for footprint. Point-turns are tighter and faster but need more traction and power.

Traction & gearing notes

- More weight over the drive wheels → more **traction** → less slipping (but slower to accelerate)
- Gear the drivetrain for your game — fast & light, or geared-down torque for pushing heavy objects
- Omni wheels trade some drive traction for strafing freedom; match wheel choice to your strategy
- Keep the chassis rigid and wheels aligned or the robot will wander off line

Drive straight: with the built-in inertial/gyro you can hold a heading so the robot tracks a straight line lap after lap (see PID under Programming).

WHY VEX IQ

WHY VEX IQ — ADVANTAGES OF THE PLATFORM

VEX IQ is built to teach real engineering and programming to younger students in a safe, fun, classroom-friendly way — and it doesn't stop at the classroom.

Advantage	What it means
Modular	Every beam, gear and plate snaps onto any other on a standard hole grid — rebuild, redesign, reuse parts across projects without tools.
Durable	Tough plastic structure survives crashes in classrooms and competitions; fewer broken parts, more learning time.
Classroom-friendly	Tool-free and low-risk for Grades 5–8; organised trays make inventory, clean-up and team management easy.
Competition ecosystem	VEX IQ Robotics Competition (VIQRC) gives every team a standard field, game objects, Robot Skills and Teamwork matches to aim for.
Real programming	Start in visual Blocks, advance to Python and C++ — the same logic used by engineers and pro-robotics teams.
Real engineering	Gears, ratios, torque, sensors, control loops and design iteration taught through hands-on builds, not lectures.
STEM pathway	VEX IQ → VEX V5 (metal) → VEX U (college) builds one continuum from age 10 to university + careers in robotics, mechatronics and software.

VOCABULARY

KEY VOCABULARY — 14 TERMS

Gear ratio — driven teeth ÷ driver teeth; sets speed vs torque of a gear train
Bevel gear — 45°-toothed gear that turns the axis of rotation by 90°
Chain — links joining sprockets to transmit rotation over a distance
Hub — fitting that locks a wheel or gear onto a shaft
Traction wheel — high-grip rubber tire for grip, climbing and pushing
PID — Proportional–Integral–Derivative control loop; corrects toward a target precisely
Chassis — the rigid base frame all components mount on
Spur gear — flat toothed wheel meshing with another on parallel shafts
Sprocket — toothed wheel that drives a chain (instead of meshing with another gear)
Shaft / axle — square/cross rod that carries rotation from a motor to gears or wheels
Omni wheel — wheel with rim rollers — can roll and also slide sideways
Encoder — device that counts rotation/motion — the Smart Motor measures to 0.375°
Drivetrain — the wheel + motor + gear system that moves the robot
Axle — a stationary or rotating cross-shaft that wheels/gears turn about